

Java Operators

The Tools for Calculation & Logical Operations

Operators are special symbols used to perform operations on variables and values. From simple addition to complex logic, they are the "verbs" of your code.

1. The Arithmetic Engine

Perform basic math calculations between values or variables.

+ (Add) | - (Subtract) | * (Multiply) | / (Divide) | % (Remainder)

```
int x = 100 + 50; // 150
int sum1 = x + 250; // 400
int finalSum = sum1 + sum1; // 800
```

2. Operator Categories

Assignment (=, +=, -=)

Assign values to variables. `x += 5` is short for `x = x + 5`.

Comparison (==, !=, >)

Compare two values. Results are always **boolean** (true/false).

Logical (&&, ||, !)

Determine logic between variables or values (AND, OR, NOT).

Bitwise (&, |, ^)

Operate on individual bits of data (Advanced usage).

3. Assignment Shortcuts

Java provides "Compound Assignment" operators to make your code more concise.

```
int score = 10;
score += 5; // score is now 15
score *= 2; // score is now 30
score /= 3; // score is now 10
```

4. Comparison & Logic

Most used in if statements to guide program flow.

```
boolean isAdult = age ≥ 18;
boolean canEntry = isAdult && hasTicket;

// && requires BOTH to be true
// || requires at least ONE to be true
```

Summary Recap

- **Arithmetic:** Perform calculations (+, -, *, /).
- **Assignment:** Stores values (=). Includes shortcuts like +=.
- **Comparison:** Evaluates equality or size. Returns true/false.
- **Logical:** Joins multiple conditions together (&&, ||).